Phase 1 — Siegert FP + Jacobian eigenvalues

`deq/closed_form_neg_loop/`, May 2026

1. Goal

For each (w_{IA}, w_{XA}) cell on the Phase 0 grid, solve the 2-population Siegert fixed point on the negative-loop topology and compute the Jacobian eigenvalues of the rate-equation linearization at that FP. Classify cells by FP stability + ringing.

2. Setup

- **Calibration** (locked from `closed_form/results/phase1_grid.npz`): $\alpha = 0.250$, $\beta = 0.00429$, $\tau_m = 2.350$, $\tau_{\text{ref}} = 0.361$, $R^2 = 0.936$.
- **Topology** (rate version): A sees external Bernoulli drive $w_{XA} \cdot p_{\text{thin}}$ ($p_{\text{thin}} = 0.7$ matches calibration) plus $w_{IA} \cdot \nu_I$; I sees $w_{AI} \cdot \nu_A$ only.
- **FP solver**: `scipy.optimize.fsolve` on the 2-D self-consistency $\boldsymbol{\nu} = \boldsymbol{\Phi}(\boldsymbol{\mu}(\boldsymbol{\nu}), \boldsymbol{\sigma}(\boldsymbol{\nu}))$, seeded from 5 initial guesses.
- **Jacobian**: $A = (1/\tau_m)(-I + \operatorname{diag}(g)J)$, with $J = \alpha\begin{pmatrix} 0 & w_{IA} \\ w_{AI} & 0 \end{pmatrix}$ and $g_i = \left(d\frac{\Phi}{d}\mu\right)|_{\text{FP}}$ (from `transfer.dphi_dmu`).

3. Closed-form structure

For this 2x2 antidiagonal J, the eigenvalues are

$$\lambda_\pm = \frac{1}{\tau_m}(-1 \pm \sqrt{g_A g_I w_{AI} w_{IA}})$$

Since $w_{IA} < 0 < w_{AI}$ and gains are positive, the radicand is **negative**, so $\lambda_\pm$ is a complex-conjugate pair with

$$\operatorname{Re}(\lambda) = -\frac{1}{\tau_m} \approx -0.426, \quad \operatorname{Im}(\lambda) = \left(\frac{1}{\tau_m}\right)\sqrt{|g_A g_I w_{AI} w_{IA}|}.$$

The FP is therefore **always a stable spiral** (Re negative, Im nonzero) unless the gains degenerate at $\nu = 0$ or $\nu = 1$. **Rate theory cannot predict sustained oscillation** — Property 5′s limit cycle lives strictly beyond mean-field reach.

4. Result

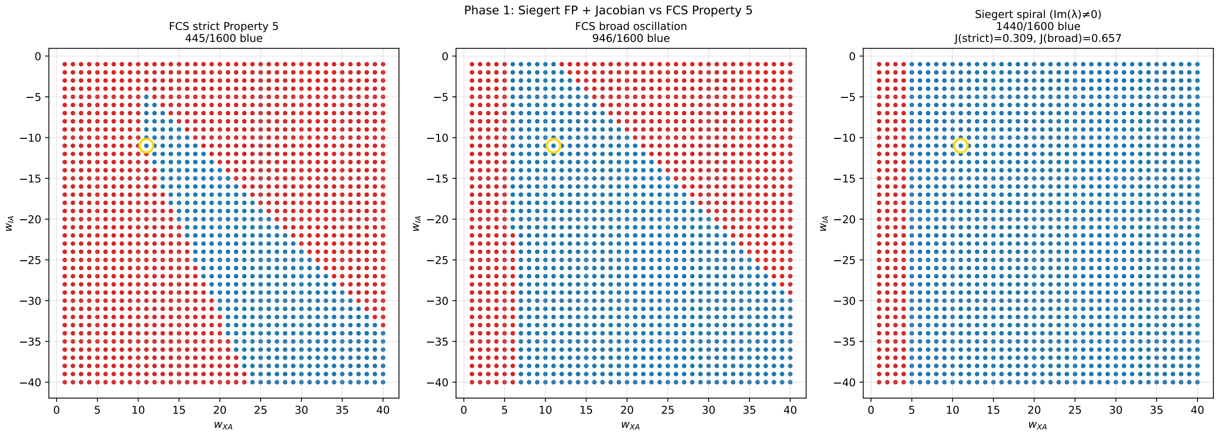

Figure 1: Three labellings on the same grid. **Left:** FCS strict Property 5 (445/1600). **Middle:** FCS broad oscillation (946/1600). **Right:** Siegert spiral-blue, where the FP exists and Im(λ) ≠ 0 (1440/1600 = 90%). The smooth-rate envelope is much wider than Property 5; almost every cell looks like a ringing spiral to mean-field theory. Gold ring: FCS default cell $(-11, 11)$.

Label	Cells	Jaccard vs Siegert spiral
Siegert spiral (Im ≠ 0)	1440 / 1600	—
FCS strict Property 5	445 / 1600	0.309
FCS broad oscillation	946 / 1600	0.657

The 0.31 Jaccard against strict Property 5 quantifies the mean-field-over-prediction: rate theory says "everywhere rings", but only 28% of cells actually oscillate at exactly period 4. Against the looser broad-oscillation label, Jaccard climbs to 0.66 — the Siegert spiral envelope is a decent **upper bound** on where oscillation can happen, but it can't pick out which specific period FCS will lock.

The 160 non-spiral cells (10%) are saturation regions: w_{XA} strongly dominates $|w_{IA}|$, A fires every tick, gains degenerate at $\nu = 1$ — Phase 1 marks these as "no spiral" because the rate equation has no analytic linearization there.

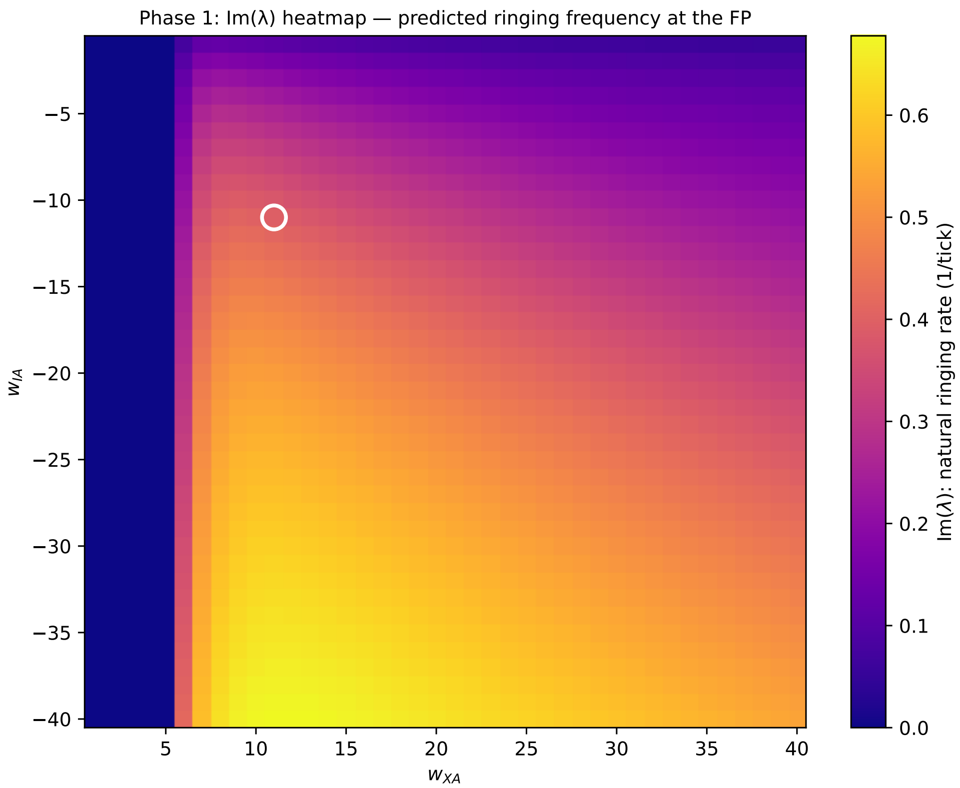

Figure 2: Heatmap of Im(λ) — the predicted ringing rate (rad / tick). Brighter is faster ringing. The ringing rate grows with both $|w_{IA}|$ and $w_{AI} \cdot w_{XA}$ (which both raise the gain product $g_A g_I w_{AI} w_{IA}$).

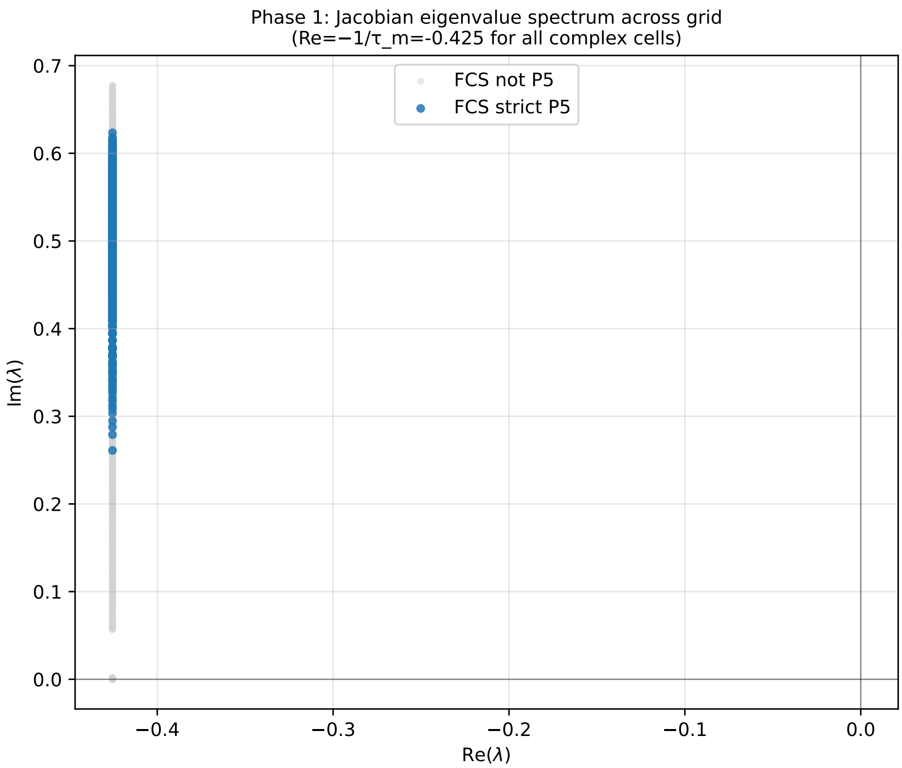

Figure 3: Jacobian eigenvalue spectrum across all converged cells. Re(λ) is locked at $-\frac{1}{\tau_m} \approx -0.426$ for every complex cell (this is the rigid prediction of single-pole low-pass H(ω)); only Im(λ) varies. Blue dots: cells where FCS strict Property 5 holds.

5. Sanity gate

FCS default cell $(w_{IA}, w_{XA}) = (-11, 11)$:

$$\boldsymbol{\nu}^\star = (0.3520, 0.1793), \quad \lambda = -0.426 \pm 0.395i$$

Stable spiral, predicted ringing period $T_{\text{pred}} = 2\pi/\ |\text{Im}(\lambda)| \approx 15.9$ ticks — about $4\times$ the FCS-measured period of 4. **Phase 1 PASS gate** (FP exists, eigenvalues complex), but Phase 2 will quantify this factor-of-4 period gap across the entire grid: the single-pole rate model predicts the **existence** of ringing but systematically **over-estimates** its period relative to discrete-tick FCS dynamics.

6. Verdict

Phase 1 PASS gate. The rate-equation FP is a stable spiral everywhere outside the saturation band; mean-field theory cannot predict Property 5′s sustained oscillation but it does provide a ringing-frequency estimate at every cell. The 0.31 Jaccard against strict Property 5 documents that smooth-rate theory has the **wrong class** of attractor here — Property 5 is a discrete-tick limit cycle, not a continuous-time Hopf bifurcation. Phase 2 turns the Im(λ) estimates into period predictions and checks them against FCS's measured period.

Output: `results/phase1/{siegert_grid.npz, hopf_vs_fcs.pdf, eig_spectrum.pdf, im_lambda_heatmap.pdf}`, `results/phase1.log`.